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# Do Text-to-Music Models Take Direction? An Instrumented Audit of Energy-Instruction Compliance

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## Abstract

Text-to-music systems are steered almost entirely through natural-language prompts, yet whether graded perceptual instructions are actually followed is largely unknown. We audit five open text-to-music models (2023–2026) on one well-measured perceptual axis: musical energy. A frozen matrix of genres  $\times$  requested levels  $\times$  phrasings (numeric “energy level  $L$  out of 11” vs. a five-word verbal ladder), six generations per cell, yields 1,500 thirty-second clips. Each clip is scored by three instruments—an interpretable, calibrated arousal model, integrated loudness (LUFS), and CLAP text–audio similarity—and every analysis is repeated on loudness-normalized copies to separate “changed the music” from “turned the volume knob.” Because the instrument is a sparse linear model over named features, compliance decomposes exactly into loudness, noisiness, and tonality—mechanisms, not only scores. We find: (1) *numeric* instructions do nothing measurable (all monotonicity CIs cover zero), while *verbal* instructions move every model (verbal–numeric  $\Delta\rho$  excludes zero for all five); (2) usable dynamic range is nonetheless narrow—median achieved movement between “very calm” and “extremely intense” spans 0.1–2.05 points on an 11-point scale; (3) a feature-level decomposition attributes compliance mostly to noisiness and tonality change rather than loudness (loudness share  $-2$ – $38\%$ ), and the one model with the largest loudness share independently loses the most compliance under loudness normalization; (4) compliance concentrates inside genre-plausible energy bands; and (5) one model’s energy *decreases* at the top instruction level on all three instruments. We release prompts, scores, features, and analysis code for full statistical reproduction.

## 1 Introduction

Prompting is the primary interface to modern text-to-music (TTM) systems, and prompts routinely contain graded perceptual direction: *a calm folk track*, *extremely intense EDM*. Whether such direction is followed—monotonically, with usable range, consistently across genres and phrasings—has, to our knowledge, not been answered with graded instruction ladders and objective measurement. Existing adherence evaluation is dominated by a single scalar, CLAP text–audio similarity to the whole prompt [1], which cannot say *which* part of a prompt was honored, *how much* the audio moved per unit of requested change, or *by what acoustic means*.

We treat this as a measurement problem. We fix one perceptual axis with unusually good measurement properties—arousal/energy, the most reliably rated affect dimension in music research [2, 3]—and deploy an interpretable, calibrated arousal instrument [4] alongside integrated loudness [5] and CLAP similarity [6] over a frozen prompt matrix. The instrument’s interpretability turns an adherence score into a *discovery*: a sparse linear model over named acoustic features, its compliance signal

decomposes exactly into loudness, noisiness, and tonality contributions—when a model “adds energy,” what did it change?

**Contributions.** (1) The first graded *compliance curves* for perceptual energy instructions across five open TTM models spanning 2023–2026 and three architecture families, with clip-level bootstrap confidence intervals for monotonicity, dynamic range, and phrasing effects. (2) A two-way mechanism decomposition—loudness-normalized re-scoring plus an exact feature-level attribution—showing that measured compliance is mostly musical-content change, quantifying each model’s “volume-knob” share, and demonstrating that the two methods agree. (3) A genre-plausibility analysis showing compliance is gated by what a genre palette can plausibly reach, and two top-of-scale phenomena: instruction saturation (energy *falling* from level 7 to level 9) and instrument ceiling effects, disentangled using pre-calibration scores. (4) A fully reproducible release: prompts, per-clip scores and features, and a notebook that regenerates every statistic in this paper from the released CSVs.

## 2 Related work

**Controllable music generation** spans text conditioning [7–9] and signal-level controls [10]; recent work adds continuous valence–arousal conditioning precisely because text-based emotional prompting is considered unreliable [11]. Our audit quantifies that assumed-weak baseline as graded curves. **Adherence evaluation** typically reports CLAP-style similarity between prompt and output [1]. Closest to our question, Almoclips [12] finds human-rated emotion conveyance biased toward neutrality; we find the objective analogue (clustered “house levels,” narrow range) with graded ladders, phrasing contrasts, and an acoustic mechanism account that human ratings cannot provide. **Music emotion recognition** supplies the measurement substrate: continuous arousal annotations [13, 14] and evidence that arousal is the stable, acoustically grounded affect axis [3].

## 3 Materials and method

**Models.** Five open checkpoints chosen to cross scale, era, and architecture: MusicGen-small and MusicGen-medium (2023; autoregressive LM over EnCodec tokens; scale axis held within family) [7]; Stable Audio Open 1.0 (2024; latent diffusion) [8]; Stable Audio 3 Small Music (2026; music-tuned latent diffusion; era axis held within vendor) [15]; and ACE-Step 1.5 turbo (2026; hybrid LM-planner+DiT, RL-tuned for prompt adherence) [9]. We abbreviate mgs, mgm, sao, sa3s, ace. Commercial systems are excluded on license-risk grounds; the roster is open-weights only by design.

**Prompt matrix (frozen before generation).** Five genre palettes (rock, EDM, hip-hop, acoustic folk, orchestral)  $\times$  five requested levels  $L \in \{1, 3, 5, 7, 9\} \times$  two phrasings: numeric (“instrumental *genre* track, energy level  $L$  out of 11”) and verbal (“instrumental *genre* track, *descriptor*” with the fixed ladder *very calm* / *calm* / *moderate energy* / *energetic* / *extremely intense*). “Instrumental” is prepended throughout to neutralize the vocal-capability confound (some rosters cannot sing). Six seeds per cell give  $5 \times 50 \times 6 = 1,500$  clips of  $\sim 30$  s.

**Instruments.** All clips pass through one front end (mono sum, 22.05 kHz, first 30 s [16]) into three scorers. (i) The Arousal Index [4]: a ridge regression over six descriptors of three named features—loudness (frame RMS), noisiness (zero-crossing rate), and a tonality index ( $-10 \log_{10}$  spectral flatness), each as mean and standard deviation—trained on DEAM arousal annotations [13] ( $R^2=0.29$ , 5-fold), transferring to PMemo [14] at Spearman  $\rho=0.77$ , and isotonically calibrated (out-of-fold) to an absolute 0–11 scale. Its known limitation is inherited honestly: DEAM under-represents arousal extremes, so the calibrated scale compresses above  $\sim 6.5$ ; we therefore also report pre-calibration (“raw”) scores where the top of the scale matters (§4.4). (ii) Integrated loudness in LUFS [5, 17]. (iii) CLAP cosine similarity between each clip and its own prompt [6]. **Normalized lens:** every clip is additionally loudness-normalized to  $-14$  LUFS and re-scored by (i) and (iii); compliance that survives normalization reflects changed musical content, not gain.

**Positive controls.** Real DEAM excerpts of known low vs. high arousal run through the identical stack score 3.7–4.1 vs. 6.0–6.5, confirming the instruments separate genuine arousal differences in this pipeline before any claim is made about generated audio.

Table 1: Monotonicity and dynamic range with 95% clip-level bootstrap CIs (B=2,000).  $\Delta$  = level-9 minus level-1 Arousal Index score (verbal); median is the pre-registered estimator, mean is the sensitivity analysis. Bold: CI excludes 0.

|      | $\rho$ numeric  | $\rho$ verbal         | $\Delta\rho$ (verb—num) | median $\Delta$          | mean $\Delta$            |
|------|-----------------|-----------------------|-------------------------|--------------------------|--------------------------|
| mgs  | .04 [−.13, .20] | <b>.39</b> [.25, .51] | <b>+.35</b> [.20, .50]  | <b>2.05</b> [0.80, 2.40] | <b>0.93</b> [0.40, 1.45] |
| mgm  | .04 [−.13, .20] | <b>.54</b> [.41, .64] | <b>+.50</b> [.33, .67]  | <b>1.30</b> [0.50, 2.10] | <b>1.28</b> [0.91, 1.63] |
| sao  | .01 [−.17, .16] | <b>.50</b> [.37, .62] | <b>+.50</b> [.37, .63]  | <b>1.30</b> [0.55, 2.00] | <b>1.13</b> [0.78, 1.50] |
| sa3s | .04 [−.14, .20] | <b>.38</b> [.22, .53] | <b>+.35</b> [.22, .48]  | <b>1.60</b> [0.55, 2.20] | <b>0.83</b> [0.31, 1.33] |
| ace  | .07 [−.09, .23] | <b>.33</b> [.17, .48] | <b>+.26</b> [.06, .45]  | 0.10 [−0.05, 0.70]       | 0.18 [−0.12, 0.47]       |

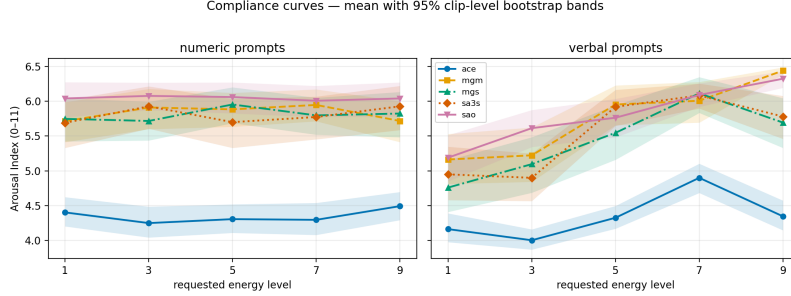


Figure 1: **Compliance curves.** Mean Arousal Index score (0–11) vs. requested energy level, per model, with 95% clip-level bootstrap bands (B=2,000), for numeric (left) and verbal (right) phrasings. Numeric curves are flat at each model’s “house level”; verbal curves rise, with model-specific slopes, ceilings, and a top-level reversal for ace. The released notebook regenerates this figure.

**Statistics.** Pre-committed before scoring: per model $\times$ phrasing, the median score per level; monotonicity as Spearman  $\rho$ (level, score); dynamic range as the level-9 minus level-1 difference; the phrasing gap; and the normalized-lens decomposition—each with 2,000-resample clip-level bootstrap percentile 95% CIs (seed-fixed; digit-reproducible from the released data). Mean-based deltas are reported alongside medians as a sensitivity analysis; where they diverge, the divergence itself is informative (§4.1). The genre-band analysis uses energy bands fixed before scoring for the four instrumentation palettes; the rock band is analyst-set from a documented prior taxonomy and flagged as such, with a rock-excluded sensitivity check.

## 4 Results

### 4.1 Numbers do nothing; words work—narrowly

Table 1 gives the headline. Under *numeric* instructions, no model moves: all Spearman CIs cover zero and every curve sits at a model-specific house level (numeric mean 6.04 sao, 5.83 mgm, 5.81 mgs, 5.80 sa3s, 4.35 ace)—an objective counterpart to the “bias toward neutrality” reported for categorical emotion intents [12]. Under *verbal* instructions every model becomes reliably monotonic ( $\rho$  from .33 to .54, all CIs excluding zero), and the verbal—numeric contrast  $\Delta\rho$  excludes zero for all five models. The instruction *format* is not cosmetic: these systems parse “extremely intense” but not “energy level 9 out of 11.”

Achieved range is another matter. The pre-registered median movement from “very calm” to “extremely intense” spans 0.10 (ace) to 2.05 (mgs) points of an 11-point scale—at best  $\sim 19\%$  of the scale. Mean-based deltas are systematically smaller for the mid-pack (mgs 0.93 vs. median 2.05; sa3s 0.83 vs. 1.60): the achieved-movement distribution is right-skewed, i.e., failure is *clip-concentrated*—a minority of generations ignore the instruction and drag the mean; the typical clip complies more than the mean suggests. For ace, neither median nor mean excludes zero: the 2026 adherence-tuned model has no established verbal energy range on this audit, in the direction opposite its design claim [9].

Table 2: Mechanism decomposition (verbal, level 9 – level 1). Content share = fraction of the calibrated  $\Delta$  surviving –14 LUFS normalization (95% CI). Cluster shares split the pre-calibration score delta  $\Delta_{\text{raw}}$  exactly; shares sum to 100% and can exceed it when clusters oppose.

|      | content share   | $\Delta_{\text{raw}}$ | loudness                            | noisiness | tonality |
|------|-----------------|-----------------------|-------------------------------------|-----------|----------|
| mgm  | .76 [.63, .91]  | +1.86                 | –2%                                 | 62%       | 40%      |
| sa3s | .82 [.64, 1.15] | +0.73                 | 18%                                 | 39%       | 43%      |
| mgs  | .76 [.47, .99]  | +0.96                 | 21%                                 | 27%       | 53%      |
| sao  | .59 [.50, .69]  | +1.39                 | 38%                                 | 38%       | 24%      |
| ace  | undefined       | +0.21                 | (no reliable $\Delta$ to decompose) |           |          |

#### 4.2 Mechanism: mostly the music, partly the volume knob

Two independent decompositions agree (Table 2). First, the *normalized lens*: the content share—the fraction of each model’s verbal  $\Delta$  that survives –14 LUFS normalization—is .76 (mgm), .76 (mgs), .82 (sa3s), and .59 (sao); ace’s ratio is undefined (its raw  $\Delta$  CI covers zero: no reliable compliance to decompose). Second, the *feature attribution*: because the Arousal Index is linear in standardized descriptors, each model’s pre-calibration score delta splits exactly into loudness, noisiness, and tonality cluster contributions. Noisiness and tonality carry the movement for every complying model; loudness contributes at most 38% (sao) and essentially nothing for mgm (–2%). The consistency check lands: sao’s feature-level loudness share (38%) matches its normalization-vanishing share ( $1 - .59 = 41\%$ ) via independent computations. Verbal LUFS deltas tell the same story from a third angle: sao raises loudness by +6.3 dB from “very calm” to “extremely intense”—an order of magnitude in power—while ace’s peak-normalizing decoder clamps loudness entirely (–0.4 dB), a design property we report as behavior, not confound. **Practitioner reading:** “adding energy” predominantly means noisier, less tonal music; sao also leans on gain, which a mastering chain would erase.

#### 4.3 Compliance is gated by genre plausibility

Requested levels were graded against per-genre plausible energy bands fixed before scoring (folk 0.5–5.5, EDM 4–8, hip-hop 2–7, orchestral 3–7; rock 5–8, analyst-set). Per-unit achieved movement concentrates on steps that stay inside the band: pooled in-band gain +0.223 per requested unit vs. +0.033 out-of-band (difference +0.190, CI [+0.096, +0.279]; per-model differences significant for ace +0.297, sa3s +0.291, mgm +0.206; not for mgs +0.132 or sao +0.026). The pattern strengthens slightly when the analyst-set rock band is excluded (pooled difference +0.217, CI [+0.111, +0.321]). Notably, the model with the largest volume-knob share (sao) is the one whose compliance is *not* band-gated—gain applies anywhere; content change is constrained by the palette. We report this as association, not causation. A genre-controlled check agrees: on a fixed-anchor CLAP axis whose anchors share each genre phrase, verbal moves every model and numeric stays flat—not cross-genre acoustics. Failure has direction: rock generations start near each model’s ceiling even at “very calm” (mgm rock level-1 mean 6.37; sa3s 6.48)—these systems cannot make *calm rock*—while folk fails upward, plateauing at 4–6 under “extremely intense.”

#### 4.4 The top of the scale: saturation vs. instrument ceiling

Three of five models score *lower* at level 9 than 7 under verbal instruction—model behavior, or our instrument’s top-end compression? Three witnesses adjudicate. For **ace** the drop is real on all three: calibrated Arousal Index –0.55 [–0.82, –0.25], own-prompt CLAP –0.086 [–0.138, –0.031], and pre-calibration score –0.37 [–0.56, –0.18]—pushed to “extremely intense,” ace produces audio that even matches its own prompt *less*. For **mgs** the calibrated drop is ambiguous (–0.42 [–0.86, +0.03]) precisely because calibration compresses the top of the scale, but the pre-calibration score drops clearly (–0.73 [–1.37, –0.15], CLAP marginal at –0.085 [–0.170, +0.004]): real saturation, masked by the instrument ceiling. For **sa3s** no witness establishes a drop. mgm and sao rise on all witnesses. The same pre-calibration lens quantifies how much the ceiling understates top-end movement (e.g., mgm verbal  $\Delta_{\text{raw}}$  = 1.86 [1.47, 2.24] vs. calibrated 1.28)—a caution for calibrated-instrument audits of generated audio. A full raw-scale replicate of Table 1 (released) preserves the ordering ( $\rho$  from

.34, ace, to .59, mgm, all CIs excluding zero), and ace’s raw  $\Delta$  (+0.21 [-0.03, +0.43]) still includes zero: the null range and the undefined content share are not ceiling artifacts.

#### 4.5 Scale, era, and the frontier

Within family, the larger MusicGen is more monotonic in point estimate (.54 vs. .39;  $\Delta\rho$  +.15 [-.00, +.32], not established). Within vendor, two years (sao→sa3s) improved musicality by ear while the compliance point estimate fell (.50 → .38;  $\Delta\rho$  -.12 [-.27, +.02]): no decline established, but the interval nearly excludes any gain—better music, not better compliance. The 2026 adherence-RL-tuned frontier model (ace), as configured here, is the weakest complier on every estimate, with a genre-specific collapse (EDM per-genre  $\rho$ =.08 vs. rock .71) and confirmed top-level saturation. A pre-registered directional hypothesis—rock shows lower verbal elasticity than instrumentation palettes—held for three of five models (rock-minus-others  $\rho$ : mgm -.21, sa3s -.27, sao -.20; mgs null; ace reversed via its EDM collapse); the edge-cell table reframes it as a ceiling: rock cannot go down.

### 5 Limitations

Our arousal instrument transfers from human-annotated music to generated audio; positive controls confirm it separates real arousal, three-instrument convergence carries the headline claims, but transfer error cannot be fully excluded—hence triangulated top-of-scale claims and reported pre-calibration scores. The calibrated scale compresses above ~6.5. Own-prompt CLAP compares each clip to a different anchor text per level; a released fixed-anchor CLAP axis (genre-matched calm/intense anchors, clap-htsat-unfused, deterministic) replicates the contrast: all five verbal  $\rho$  CIs exclude zero (.27-.60); numeric flat. CLAP inference was not bit-reproducible across runs (cell-mean drift  $\leq 0.033$ ; Arousal Index and LUFS reproduce exactly); all CLAP verdicts are stable across observed draws; released values are from the canonical run. Six seeds per cell bound precision (hence CIs everywhere); clips are single 30-s windows; prompts are English; five genre palettes; open models only; no listener study—human benchmarks [12] are complementary evidence.

### 6 Conclusion

Deployed text-to-music systems take perceptual direction only in words, move a fraction of the requested range, mostly change the music rather than the volume—except where they don’t, quantifiably—and comply mainly where the genre palette makes the request plausible. A calibrated, interpretable instrument makes each claim checkable: its weights decompose compliance. Prompts, scores, features, and the statistics notebook are released at <https://github.com/ConnerFarrell/energy-audit>.

**Responsible use statement.** This work audits open-weights music generators with open data; findings are aggregate measurements, not product judgments, and the released multi-instrument stack exposes single-metric gaming. Audio exhibits respect each model’s license (MusicGen CC-BY-NC; Stability attribution; ACE-Step MIT); no human subjects or personal data are involved. Residual risks—steering generation toward manipulative affect, or misreading the instrument as a general emotion proxy—are mitigated by releasing the full stack, its negative results, and calibration limits.

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## A Per-cell means and edge grading

Full model  $\times$  genre  $\times$  level verbal means with out-of-band markers are released as `s6_edge_cells.csv`; Table 3 reproduces the two directional examples cited in §4.3.

Table 3: Directional failure examples (verbal mean Arousal Index score; \* = requested level outside the genre band).

|            | $L=1$ | $L=3$ | $L=5$ | $L=7$ | $L=9$ |
|------------|-------|-------|-------|-------|-------|
| mgm / rock | 6.37* | 6.38* | 6.48  | 6.50  | 6.48* |
| mgm / folk | 4.43  | 4.03  | 5.82  | 4.90* | 6.25* |

## B Native-unit feature deltas

Level-9 minus level-1 changes in native units (verbal), released in `s6_feature_decomp.csv`: e.g., *mgm* raises zero-crossing rate by +0.126 and *lowers* the tonality index by  $-14.8$  dB-equivalent while its RMS loudness is flat ( $-0.009$ ); *sao* raises RMS by +0.106 (the largest in the roster), consistent with its 38% loudness share.

## C Reproducibility

Corpus: 1,500 clips,  $\sim 30$  s, six fixed seeds per cell. Scoring front end identical for all models (mono, 22.05 kHz, 30 s). All bootstrap CIs: 2,000 clip-level resamples, percentile method, fixed seed; every statistic in the paper regenerates digit-for-digit from the released `scored_full_v2.csv` (per-clip scores, both lenses), `features_verbal.csv` (per-clip descriptors), and the `s6_*.csv` and `s7_*.csv` banks via the released notebook. Model checkpoints and exact revisions are listed in the repository.

## D Prompt set

All 50 prompts follow the two templates of §3; the full list ships as `prompts.csv`.